Experiment

A1-1

English

Absorption spectroscopy and chemical kinetics

Part A. Calibration (1.7 points)

A.1 (0.1pt)

Spectrum saved: LED-CON.csv

A.2 (0.1pt)

Background saved: dark-A.csv

A.3 (0.3pt)

Spectra saved: R-0.csv G-0.csv B-0.csv

A.4 (0.8pt)

LED	$\lambda_{\max}$, nm	N
Red		
Green		
Blue		

$a =$ $b =$

A.5 (0.4pt)

$\lambda_{\max} =$

Part B. Beer–Lambert law (5.0 points)

B.1 (1.0pt)

$V_d =$

B.2 (0.2pt)

Spectra saved: src-B.csv dark-B.csv

B.3 (0.4pt)

Wavelength range:

B.4 (1.0pt)

Drops	File

B.5 (0.4pt)

$\beta(\lambda) =$

B.6 (0.4pt)

$I_\lambda =$

B.7 (1.4pt)

λ, nm	$\sigma(\lambda)$

Plot:

B.8 (0.2pt)

$\lambda_{\max} =$

Part C. Phenolphthalein (6.7 points)

C.1 (0.1pt)

$J =$

C.2 (0.3pt)

Equation:

C.3 (0.3pt)

Equation:

C.4 (0.2pt)

Cuvette	NaOH, ml	NaCl, ml	Water, ml	c, mol/l

C.5 (1.5pt)

Measurements done for all five solutions:

C.6 (0.3pt)

$\lambda_{\mathrm{max,In}} =$

C.7 (1.5pt)
Plot:

C.8 (2.0pt)
Plot:

C.9 (0.5pt)

$p =$

Part D. Chemical traffic light (6.6 points)

D.1 (0.4pt)

Colour	Order	t, s
Green / cyan		
Yellow		
Red / orange		
Violet		

D.2 (1.0pt)

Spectra saved: src-D.csv dark-D.csv time series

D.3 (1.2pt)
Plot:

$\beta_1(\lambda),\ \beta_2(\lambda)$	$\beta_3(\lambda),\ \beta_4(\lambda)$

D.4 (1.0pt)

Number of states:

D.5 (3.0pt)
Plot: